

Forecasts by Market Regime

One figure for every cell of Table 7: four forecast horizons crossed with three market conditions, on CBA.AX.

What these figures show

Table 7 in the paper reports a single error number for each combination of forecast horizon and market condition. These twelve figures show the forecasts those numbers came from, so the behaviour behind each cell can be seen rather than inferred.

1. The task, in plain terms

The data is the daily closing price of one share, Commonwealth Bank of Australia, from January 2020 to July 2023. That is about 900 trading days. The question is whether a model can look at recent prices and say something useful about future ones.

Each model is given the closing prices of the last 20 trading days, roughly a month of trading. From those 20 numbers it produces the next h numbers in a single shot, where h is the forecast horizon. At $h = 1$ it produces tomorrow's price. At $h = 200$ it produces the next two hundred days all at once, rather than predicting one day and feeding it back in.

The 20-day window then slides forward by one day and the whole thing happens again. Repeating this across the test period gives several hundred separate forecasts, each made from a different starting point. Those are the individual points in the figures.

Crucially, the models are only ever tested on stretches of time that came after the data they were trained on. A model is never asked about a period it has already seen.

2. The two models being compared

LSTM is a long established type of neural network built for sequences. It reads the 20 days in order, carrying a memory of what it has seen so far, which in principle suits price data.

KAN, a Kolmogorov-Arnold Network, is a newer design. Rather than reading the days in sequence, it receives all 20 days at once as a flat list of numbers and learns a flexible curve for each input. It has no built-in notion of order, which is why the original expectation was that it would do worse on this kind of data.

3. The naive baseline, and why it matters

Alongside the two models there is a third line on every figure, and it is not a model at all. The **naive baseline** simply predicts that the price will not change. Whatever today's closing price is, that is its forecast for tomorrow, for next week, and for two hundred days from now. It does no learning and has no parameters.

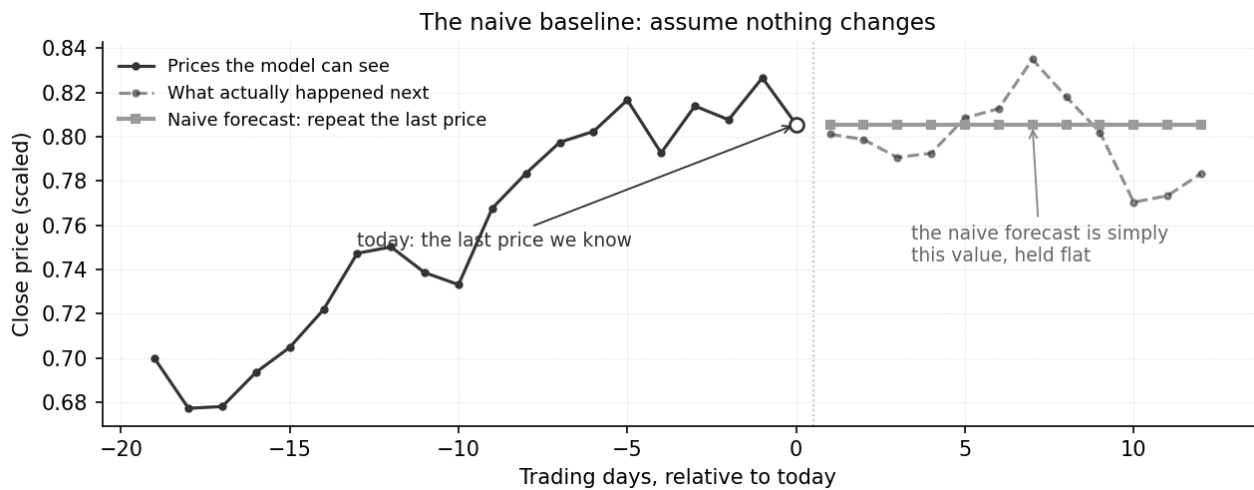


Figure A. The naive baseline in action. It takes the last known price and holds it flat for the whole forecast period.

This sounds too crude to be worth plotting, but it is the standard yardstick in forecasting, for a simple reason: share prices move slowly from one day to the next, so "nothing changes" is already a decent guess. A model that cannot beat it has not learned anything useful about the market, however sophisticated it looks. Reporting a low error without this comparison can make a model appear far more capable than it is.

Including it is one of the things that changed in the revised analysis, and it is the single most useful line on these plots for judging whether either model is genuinely forecasting.

4. The three market conditions

The point of Table 7 is that a model might cope well in calm markets and badly in turbulent ones. To test that, every 20-day window is sorted into one of three conditions before any forecast is judged.

The sorting uses only the 20 days the model can actually see. No information from the forecast period is used, so a window is never labelled using knowledge of what happened next. Two things are measured: how much the price jumped about from day to day, and how directional the movement was, meaning whether it travelled steadily one way or wandered up and down and ended where it started.

Volatile: day-to-day movement in the top quarter of what was seen during training. **Trending:** movement not unusually large, but unusually one-directional, again in the top quarter. **Normal:** everything else, and the most common case.

The cut-offs come from the training data only, so they are fixed before the test period is looked at.

Real examples of each market condition, from the data

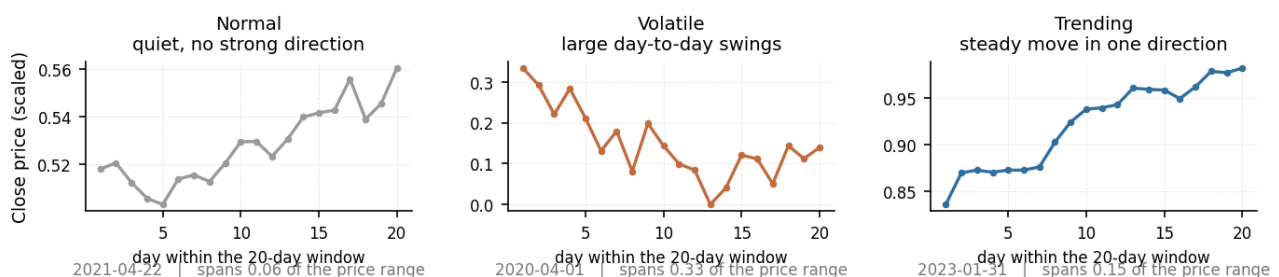


Figure B. One real window of each type, taken from the data. Note the vertical scales differ: the Volatile example covers a third of the stock's entire price range in a single month, which is why it counts as volatile.

5. Reading the twelve figures

The vertical axis is the closing price scaled from 0 to 1, where 0 is the cheapest day in the whole period and 1 the dearest. These are not dollars. Scaling was applied before training and every error figure in the paper uses the same scale, so an error of 0.05 means a typical miss of about five percent of the stock's full price range.

The horizontal axis counts the windows belonging to that regime, in chronological order. It is not a calendar. Filtering to one condition leaves gaps: a volatile stretch in March, another in June, and quiet weeks in between that belong to a different condition. Drawing a continuous calendar line across those gaps would imply a continuity that does not exist. Tick labels give the real date at each position, so a large jump between two ticks means the intervening period was a different condition.

The lines. Black is what actually happened. Blue is the LSTM. Orange is the KAN. Grey is the naive baseline. Markers show the individual forecasts. Where a coloured line sits on the black one, that forecast was right; the vertical gap is the error.

The error figures quoted beneath each plot are root mean square errors: roughly, the typical size of that vertical gap. Lower is better.

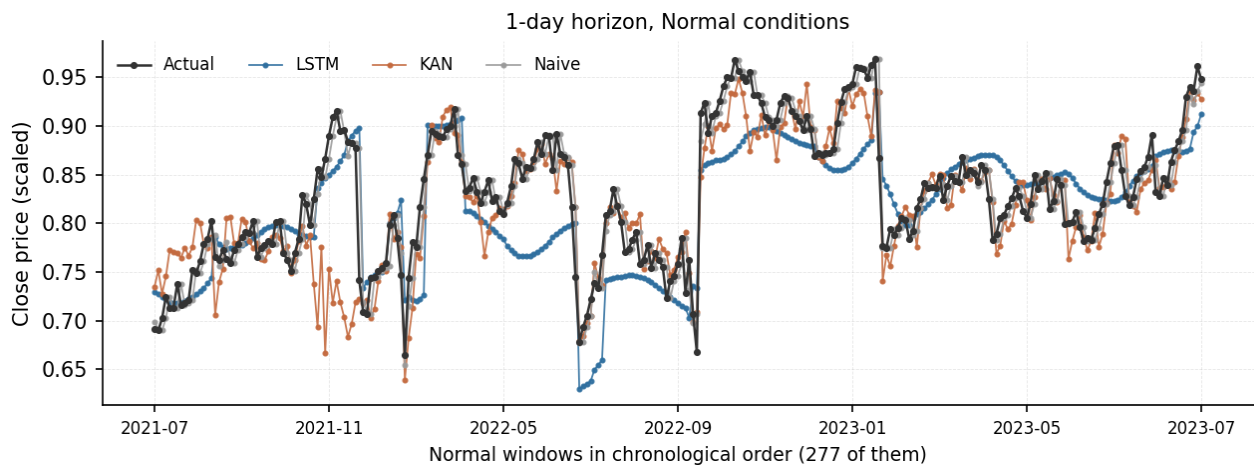
6. How these relate to Table 7

The window counts match Table 7 exactly, which is a useful check that the same data underlies both. At the 1-day horizon, for instance, both give 277 Normal, 175 Volatile and 79 Trending windows.

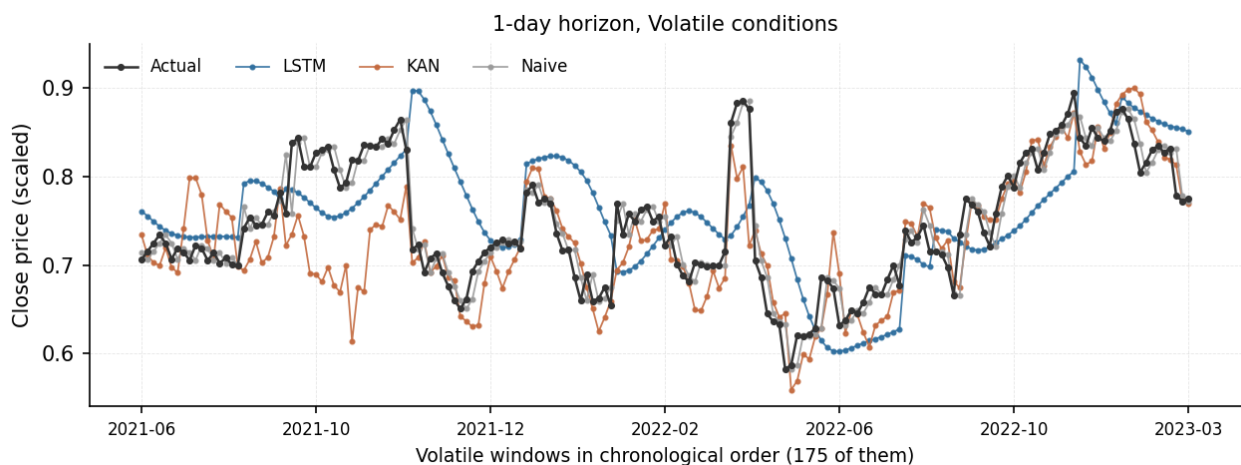
The error figures, however, are not the Table 7 numbers, and should not be quoted as though they were. Table 7 averages over every step of the forecast and over three random seeds. These figures plot only the first step of each forecast, from a single seed, because a hundred overlapping lines cannot be drawn legibly. The one-step-ahead component is the easiest part of any forecast, so the errors here are smaller than Table 7 throughout, and the naive baseline looks far stronger than it does in the paper. Use these figures to see *how* the models behave in each regime; use Table 7 for *how well*.

1-day horizon

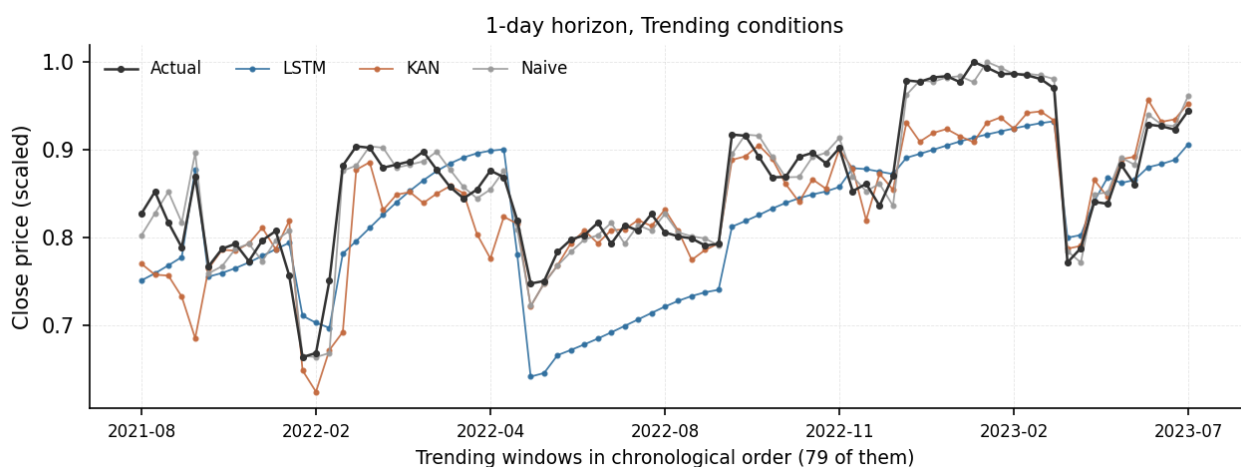
One day ahead. All three methods stay close to the actual price, since tomorrow's price is rarely far from today's.



Normal conditions: 277 windows, 2021-07-16 to 2023-07-28. Error on these lines: LSTM 0.046, KAN 0.044, naive 0.020.



Volatile conditions: 175 windows, 2021-06-23 to 2023-03-13. Error on these lines: LSTM 0.070, KAN 0.053, naive 0.020.



Trending conditions: 79 windows, 2021-08-09 to 2023-07-27. Error on these lines: LSTM 0.065, KAN 0.048, naive 0.019.

Summary for this horizon

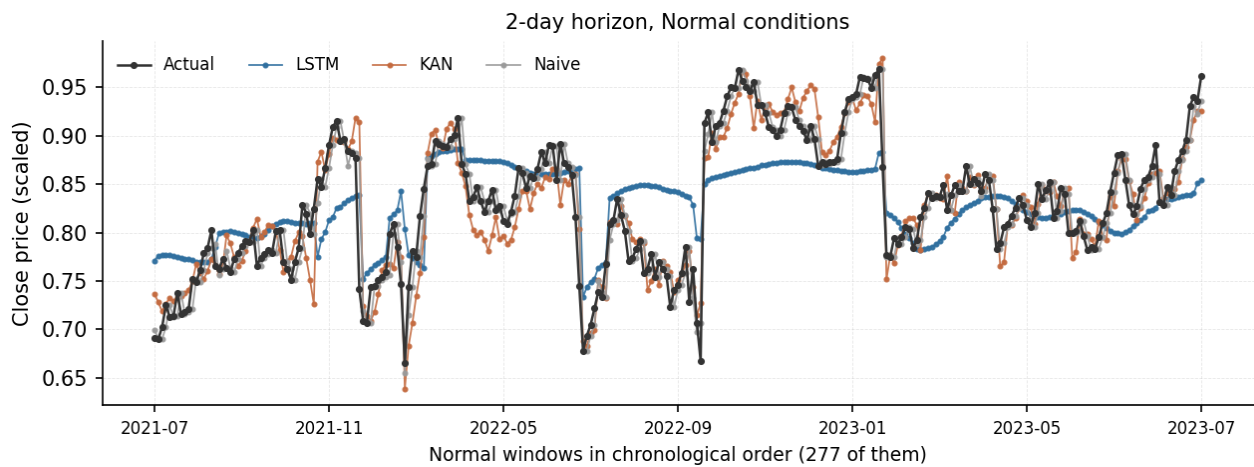
Regime	Windows	LSTM	KAN	Naive
Normal	277	0.046	0.044	0.020
Volatile	175	0.070	0.053	0.020
Trending	79	0.065	0.048	0.019

The KAN is slightly closer than the LSTM in all three regimes. Both degrade in Volatile conditions, the LSTM more so, which is the pattern Table 7 reports.

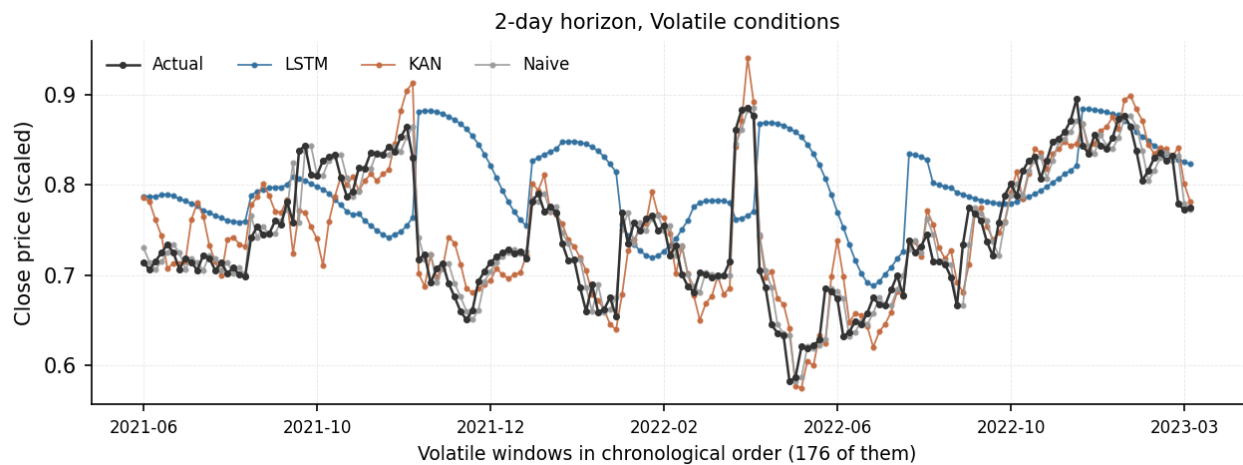
The naive baseline is the strongest line on all three plots. At one step ahead this is expected and is not by itself evidence against the models; the fuller comparison in Table 7 is the fair one.

2-day horizon

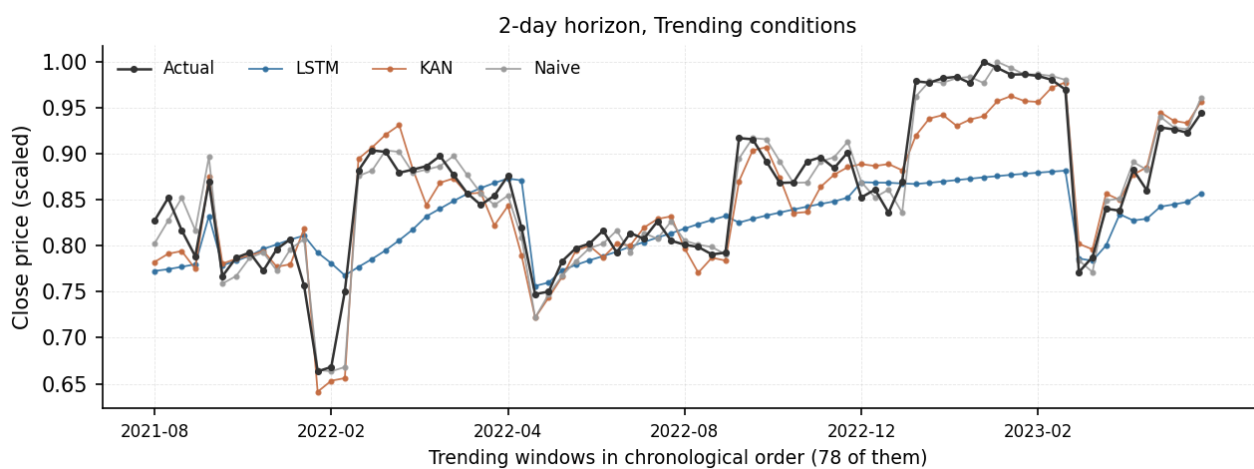
Two days ahead. The clearest separation between the two models anywhere in this set.



Normal conditions: 277 windows, 2021-07-16 to 2023-07-26. Error on these lines: LSTM 0.050, KAN 0.028, naive 0.020.



Volatile conditions: 176 windows, 2021-06-22 to 2023-03-13. Error on these lines: LSTM 0.099, KAN 0.034, naive 0.020.



Trending conditions: 78 windows, 2021-08-09 to 2023-07-27. Error on these lines: LSTM 0.064, KAN 0.030, naive 0.019.

Summary for this horizon

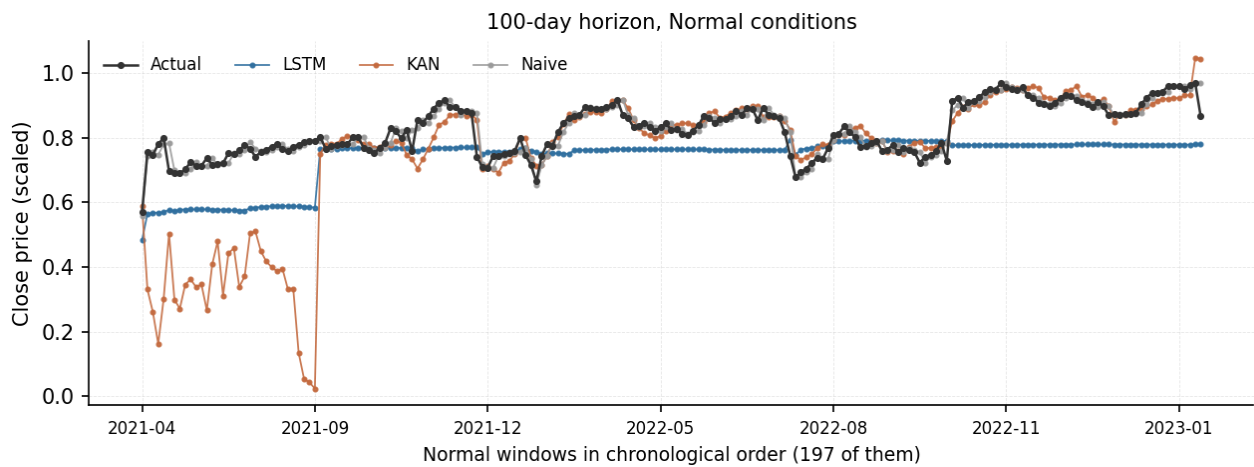
Regime	Windows	LSTM	KAN	Naïve
Normal	277	0.050	0.028	0.020
Volatile	176	0.099	0.034	0.020
Trending	78	0.064	0.030	0.019

The LSTM produces a smooth, slow-moving line that misses most turning points, and in Volatile conditions it swings far away from the actual price for weeks at a time. The KAN tracks the real movement closely in every regime.

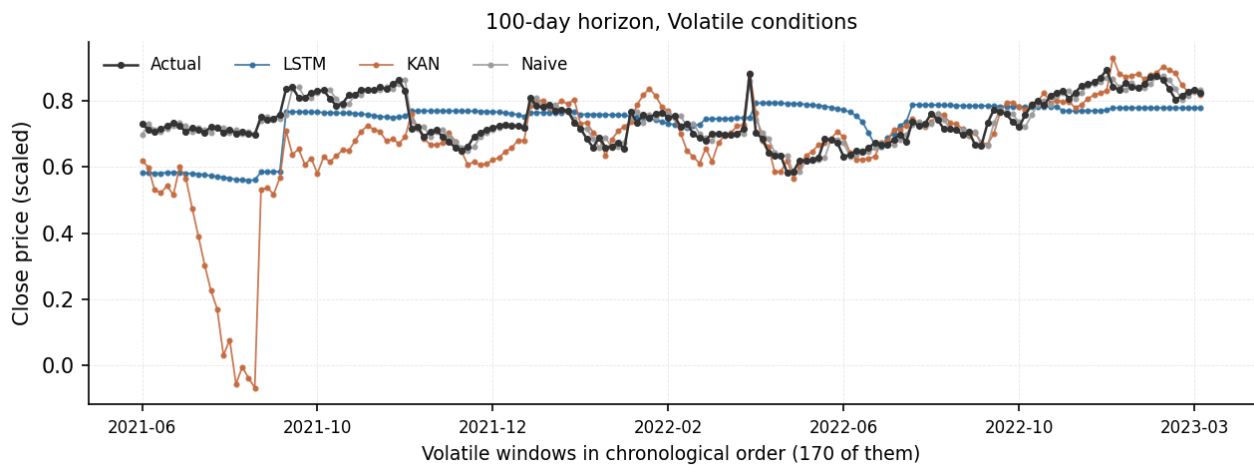
This is the horizon at which the difference between the two architectures is most visible to the eye rather than only in a table.

100-day horizon

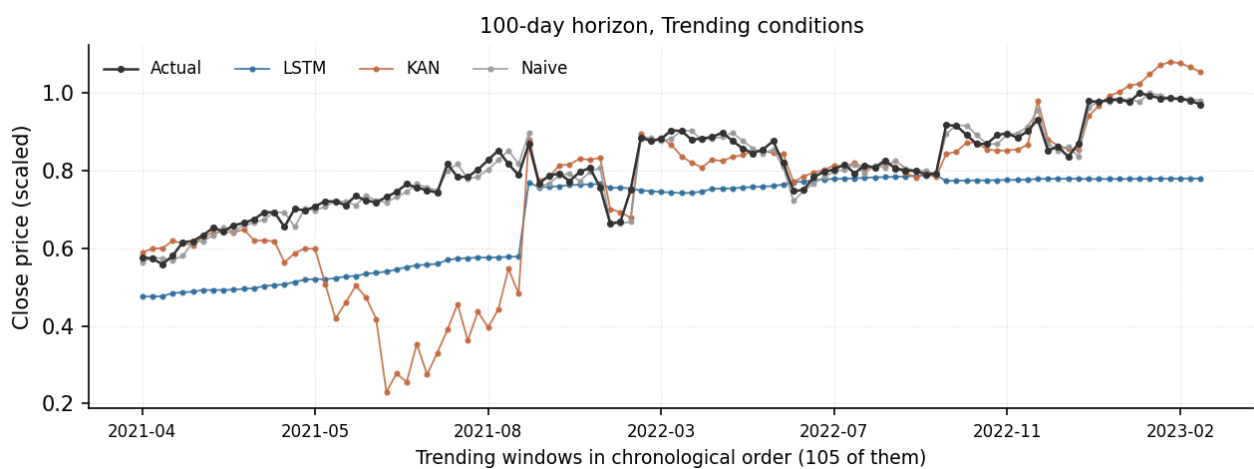
One hundred days ahead. The ordering reverses here.



Normal conditions: 197 windows, 2021-04-30 to 2023-02-14. Error on these lines: LSTM 0.115, KAN 0.181, naive 0.023.



Volatile conditions: 170 windows, 2021-06-21 to 2023-03-07. Error on these lines: LSTM 0.089, KAN 0.174, naive 0.020.



Trending conditions: 105 windows, 2021-04-27 to 2023-02-10. Error on these lines: LSTM 0.147, KAN 0.169, naive 0.019.

Summary for this horizon

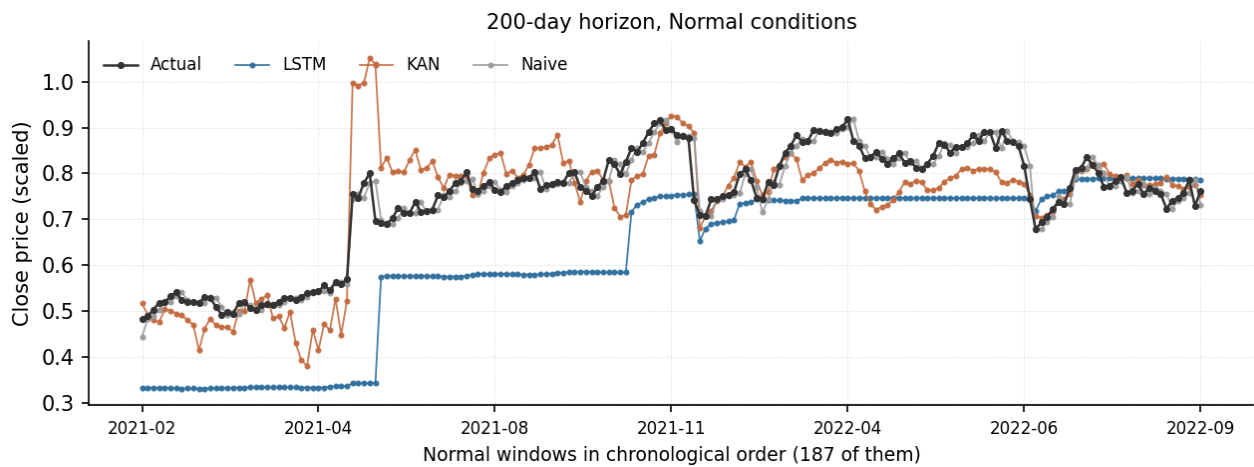
Regime	Windows	LSTM	KAN	Naïve
Normal	197	0.115	0.181	0.023
Volatile	170	0.089	0.174	0.020
Trending	105	0.147	0.169	0.019

The LSTM is the better of the two models in all three regimes, which matches Table 7. The KAN wanders further from the actual price, particularly in Normal conditions.

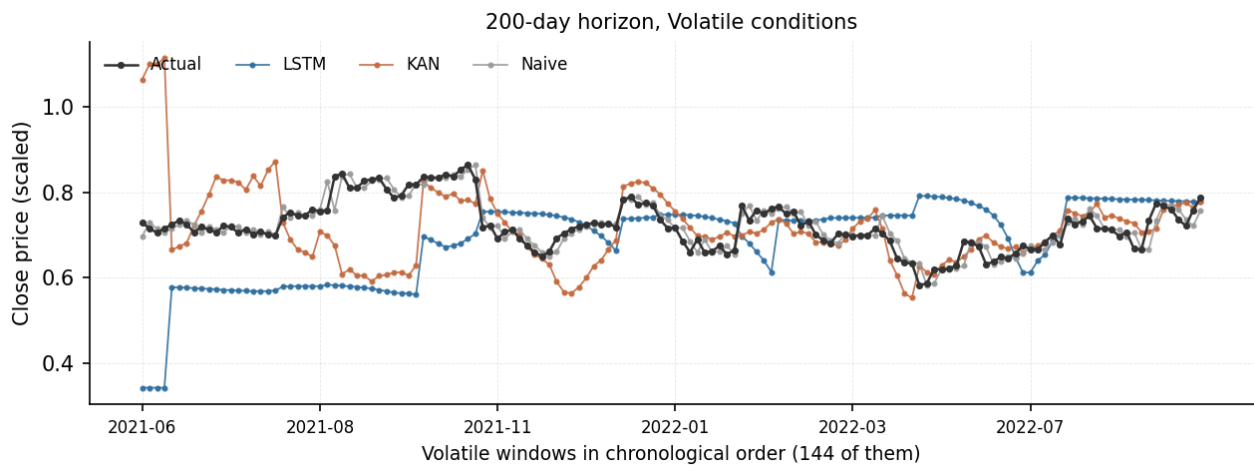
Both are now far enough from the truth that neither is tracking the price in any useful sense. The lines drift around the general level rather than following it.

200-day horizon

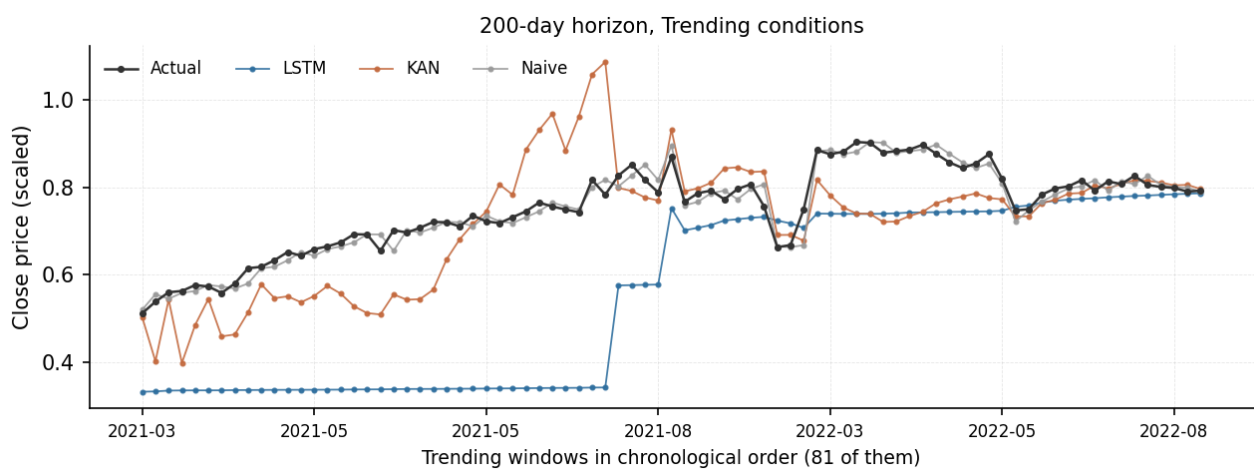
Two hundred days ahead. The ordering reverses once more.



Normal conditions: 187 windows, 2021-02-26 to 2022-09-19. Error on these lines: LSTM 0.158, KAN 0.074, naive 0.022.



Volatile conditions: 144 windows, 2021-06-21 to 2022-10-12. Error on these lines: LSTM 0.133, KAN 0.103, naive 0.021.



Trending conditions: 81 windows, 2021-03-25 to 2022-08-12. Error on these lines: LSTM 0.243, KAN 0.106, naive 0.020.

Summary for this horizon

Regime	Windows	LSTM	KAN	Naïve
Normal	187	0.158	0.074	0.022
Volatile	144	0.133	0.103	0.021
Trending	81	0.243	0.106	0.020

The KAN is closer than the LSTM in all three regimes, and the LSTM is at its worst in Trending conditions, where it is roughly twice as far from the actual price as the KAN.

The instability noted in the paper is visible here: at this horizon the KAN's optimiser struggles on the earliest and smallest training fold, which is why the paper reports those figures separately.

What the twelve figures show together

Neither model is uniformly better. The KAN is ahead at 1, 2 and 200 days; the LSTM is ahead at 100 days. Which one leads depends on the horizon rather than on the market condition, and no regime reverses the ordering within a horizon.

Volatile conditions are harder for both models at short horizons, and the LSTM suffers more from them than the KAN does. At long horizons the pattern changes: Trending windows become the difficult ones, particularly for the LSTM at 200 days.

Across every figure the LSTM produces smoother lines than the KAN. That is not a virtue in itself. A smooth line that lags the market is simply a slow-moving average, and much of the LSTM's error at short horizons comes from arriving at each turn too late.

Reproducing these

Run `make_regime_figures.py` to regenerate all twelve images and their statistics, then `build_regime_report.py` to rebuild this document. The figures are drawn from the same logged prediction files that produce Table 7, pooling all five walk-forward folds at seed 0.